

Team Name : AstroFusion

Team Leader Name : Aniruddh Parmar

Problem Statement : Optimizing Urban Heat Mitigation and cooling strategies via Artificial Intelligence and Machine Learning (AIML)

Team Members

Team Leader:

Name : Aniruddh Parmar
College : Darshan University

Team Member-1:

Name : Darpan Parmar
College : Darshan University

Team Member-2:

Name : Vishal Baraiya
College : Darshan University

Team Member-3:

Name : Meet Pipaliya
College : Darshan University

- How different is it from any of the other existing ideas?

Existing Solutions	Our AI-Based Solution
Display heat maps and satellite imagery only.	Predicts **future heat risk** using AI models.
Show **where** hotspots exist.	Explains **why** hotspots occur using NDVI, NDBI, NDWI, weather, and elevation.
Provide only static visualization.	Performs intelligent **root-cause analysis** of urban heat.
No mitigation planning.	Recommends **location-specific cooling strategies**.
No impact estimation.	Estimates the **expected temperature reduction** for each strategy.
Require expert interpretation.	AI chatbot provides instant insights and decision support.

- How will it be able to solve the problem?

Step 1: Collect Satellite Data

- We automatically collect multi-season satellite and environmental data using **Google Earth Engine**. The data includes Land Surface Temperature (LST), NDVI (vegetation), NDBI (built-up areas), NDWI (water bodies), Elevation, and geographical coordinates. Multiple months and years are used so the model learns seasonal and long-term climate patterns instead of relying on a single snapshot.

Step 2: Build a Training Dataset

- The collected data is exported as CSV files and merged into a single training dataset. Data preprocessing is performed by removing missing values, handling outliers, normalizing features, and preparing the data for machine learning.

Step 3: Train the AI Model

- The processed dataset is used to train machine learning models that learn the relationship between environmental factors and Land Surface Temperature. The trained model can accurately identify heat prone areas and estimate future temperature trends.

Step 4: Detect Heat Hotspots

- The trained model generates high-resolution heat maps and identifies Urban Heat Island hotspots across different cities and regions of India. These hotspots are ranked according to their heat risk.

Step 5: Identify the Causes

Unlike conventional heat maps, our platform explains *why* an area is experiencing high temperatures by analyzing factors such as:

- Low vegetation cover (NDVI)
- High built-up density (NDBI)
- Lack of water bodies (NDWI)
- Terrain characteristics (Elevation)
- Seasonal weather patterns
- This provides decision-makers with the root causes of urban heating.

Step 6: Recommend Cooling Strategies

Based on the identified causes, the platform generates location-specific recommendations such as:

- Tree plantation
- Urban forests
- Green roofs
- Cool roofs
- Water body restoration
- Increased green spaces
- For each recommendation, the system estimates the expected reduction in temperature through scenario-based simulations.

Step 7: AI Decision Support

- An integrated AI chatbot allows users to ask questions such as:
- Why is this location hotter?
- What is causing the hotspot?
- Which mitigation strategy is most effective?
- How can this area’s temperature be reduced?
- The chatbot answers using the trained model, satellite-derived features, and generated heat maps.

Final Impact

Our platform transforms raw satellite observations into actionable intelligence. Instead of only identifying heat hotspots, it predicts future heat risk, explains the environmental causes, and provides scientifically informed mitigation strategies. This enables governments, urban planners, and Smart City authorities to make proactive, data-driven decisions for reducing urban heat and improving climate resilience.

Unique Selling Proposition (USP)

UrbanHeat AI is not just a heatmap generation tool; it is a complete AI-powered urban climate decision support platform that transforms satellite data into actionable solutions.

1. End-to-End AI Solution

- Unlike traditional systems that only visualize temperature, our platform automates the complete workflow from satellite data collection to prediction, analysis, and mitigation recommendations.

2. Multi-Source Satellite Intelligence

- The platform integrates multiple geospatial datasets including Land Surface Temperature (LST), NDVI, NDBI, NDWI, Elevation, and seasonal information collected through Google Earth Engine, providing a comprehensive understanding of urban heat dynamics.

3. AI-Based Heat Prediction

- Instead of only displaying current conditions, the system predicts future urban heat patterns using machine learning trained on multi-season and multi-year satellite observations.

4. Explainable AI

- The platform identifies the environmental factors responsible for heat hotspots, such as low vegetation, dense built-up areas, lack of water bodies, and terrain characteristics, making predictions transparent and easy to interpret.

5. Intelligent Cooling Recommendations

- The system recommends location-specific mitigation strategies such as tree plantation, green roofs, cool roofs, and water body restoration based on the characteristics of each hotspot rather than providing generic advice.

6. Interactive AI Assistant

- The platform includes an AI assistant that allows users to ask questions in simple language about heat hotspots, future temperature predictions, and cooling strategies. It provides quick, data-driven answers using satellite data, AI model predictions, and heat map analysis.

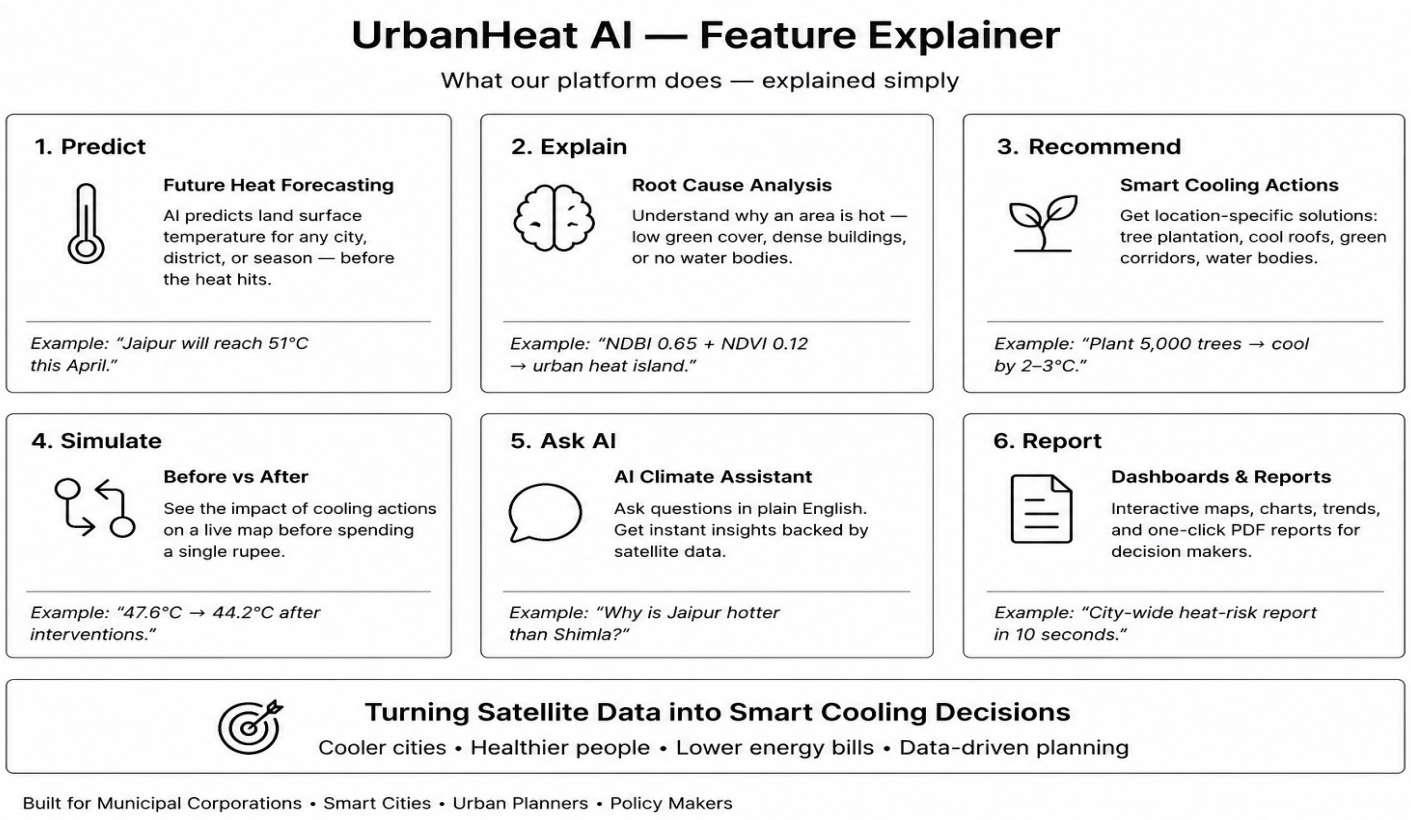
7. Decision Support for Urban Planning

- The system helps municipal corporations, Smart City authorities, and urban planners identify high-risk areas, prioritize mitigation measures, and make informed decisions for sustainable urban development.

8. Scalable and Dynamic Platform

- Built on Google Earth Engine, the platform can be easily updated with the latest satellite data. This enables continuous monitoring and makes the solution suitable for deployment across multiple cities and regions of India.

➢ List of features offered by the solution



1. Automatic Satellite Data Collection

- Our platform automatically collects satellite data using **Google Earth Engine**, eliminating the need for manual downloads. It gathers information from multiple seasons and years to ensure reliable and up-to-date analysis.

2. Multi-Source Environmental Analysis

- Instead of relying on a single dataset, our solution combines multiple environmental factors such as vegetation, built-up areas, water bodies, elevation, and land surface temperature. This helps the model understand the real reasons behind urban heating.

3. AI-Based Heat Prediction

- The collected data is used to train an AI model that predicts heat distribution across different regions. The model learns from historical patterns and environmental conditions to identify areas that are more vulnerable to heat.

4. Interactive Heat Maps

- The platform generates clear and interactive heat maps for India, allowing users to explore heat conditions at the state, district, or city level. Users can zoom, filter, and compare heat patterns across different seasons.

5. Heat Hotspot Detection

- UrbanHeat AI automatically detects heat hotspots and identifies high-risk areas that need immediate attention. This helps authorities focus their efforts where they can achieve the greatest impact.

6. Smart Cooling Recommendations

- Instead of only identifying the problem, the platform also suggests practical solutions. Based on the conditions of each location, it recommends measures such as tree plantation, green roofs, cool roofs, water body restoration, and increasing green spaces. It also estimates the expected reduction in temperature for each recommendation.

7. AI Climate Assistant

- The platform includes an AI-powered chatbot that allows users to ask questions in simple language. It can explain heat maps, identify the causes of urban heat, compare different locations, and recommend the most suitable mitigation strategies based on the available data.

8. Real-Time Dashboard

- A clean and user-friendly dashboard brings together heat maps, AI predictions, trends, charts, and recommendations in one place. This makes it easier for planners, researchers, and decision-makers to understand the overall situation quickly.

9. Automatic Report Generation

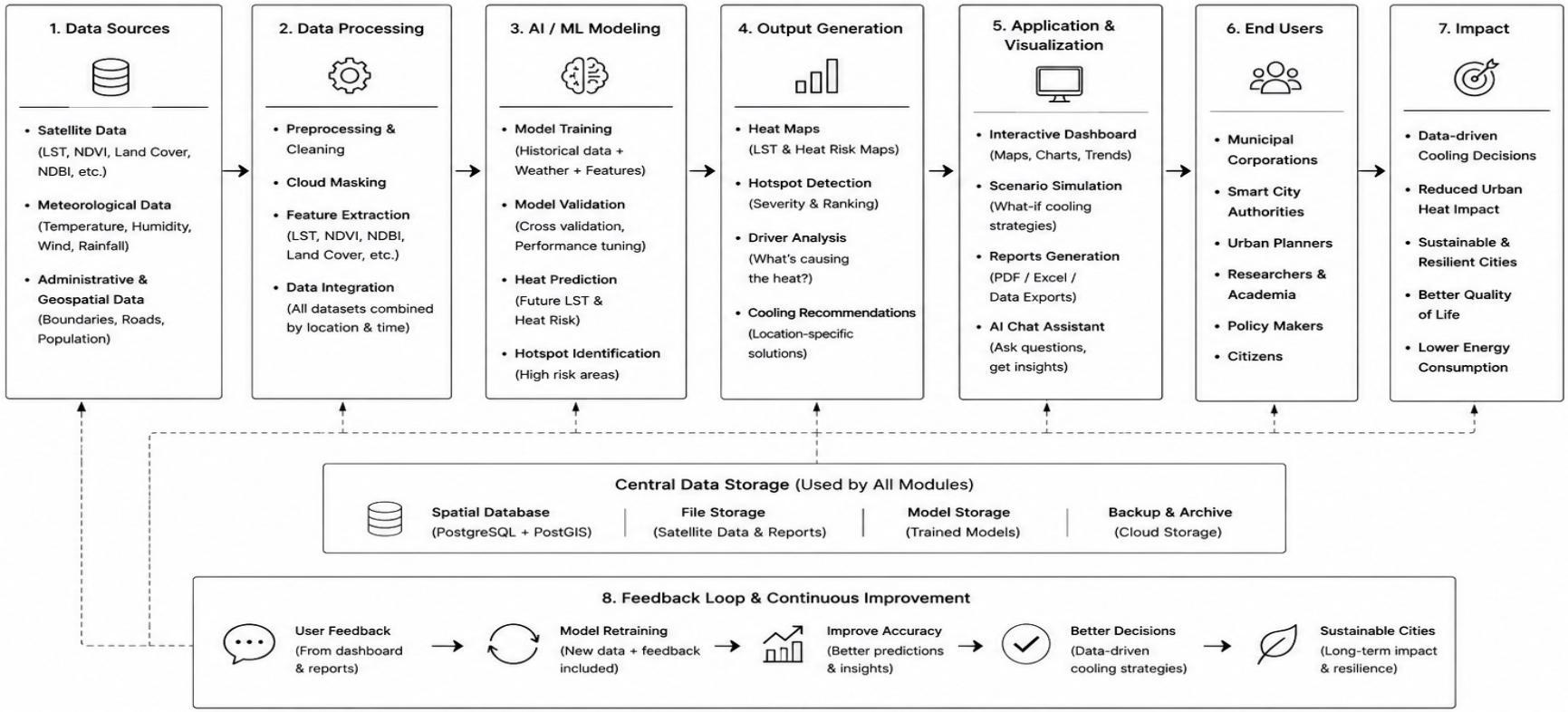
- Users can generate detailed reports with a single click. These reports include heat maps, hotspot analysis, AI predictions, and suggested mitigation strategies, making them useful for planning and decision-making.

10. Decision Support for Smart Cities

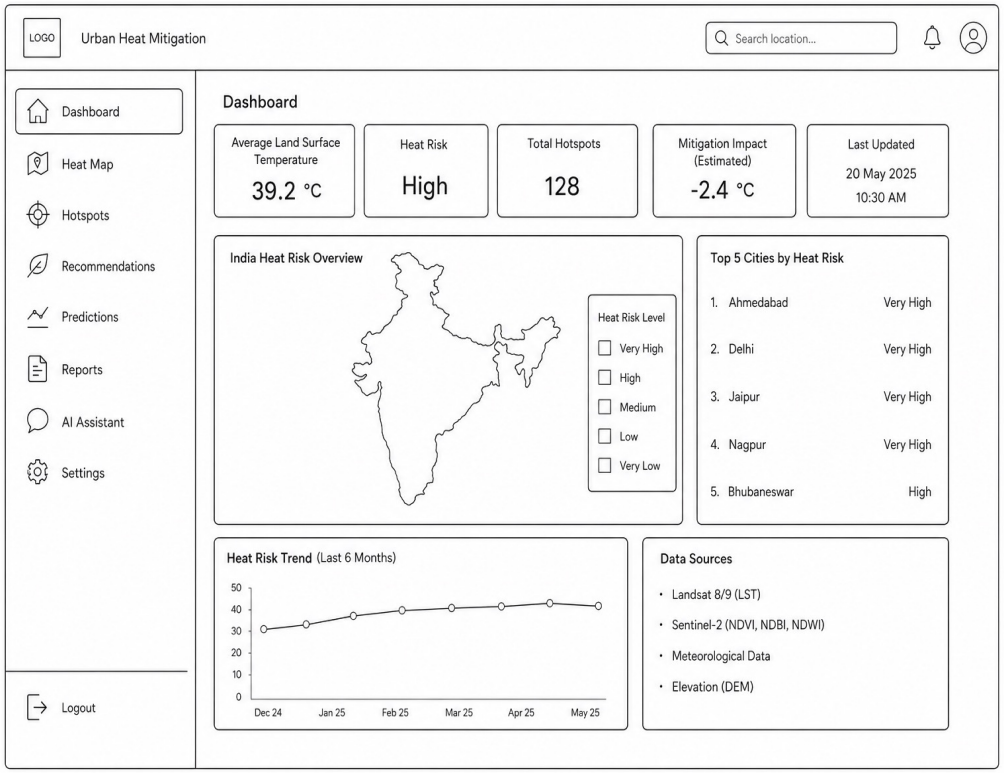
- Our platform is designed to assist municipal corporations, Smart City authorities, urban planners, and environmental agencies by providing data-driven insights that support better planning and climate-resilient development.

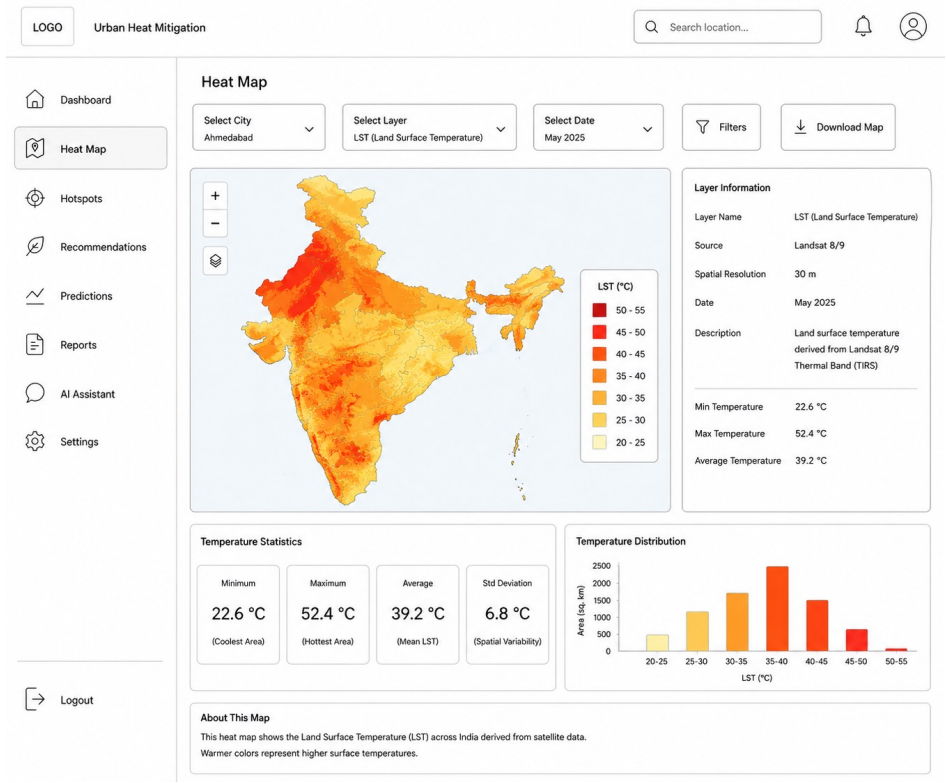
UrbanHeat AI — Process Flow

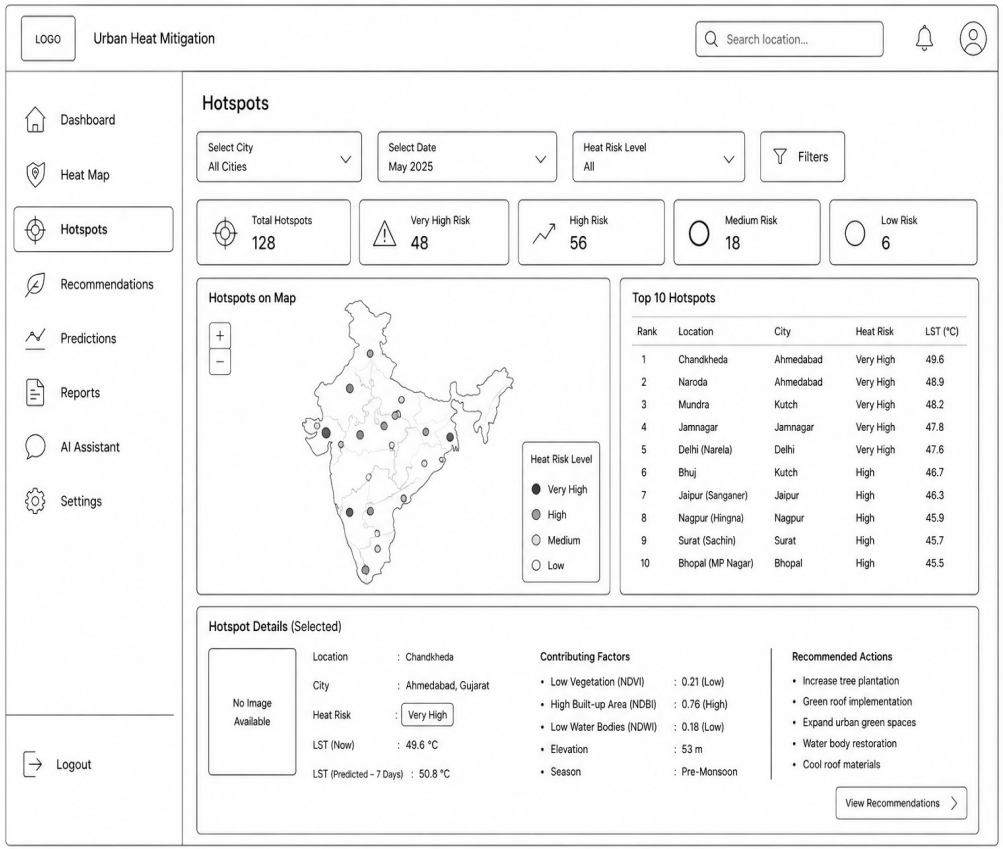
From satellite data to smart cooling recommendations

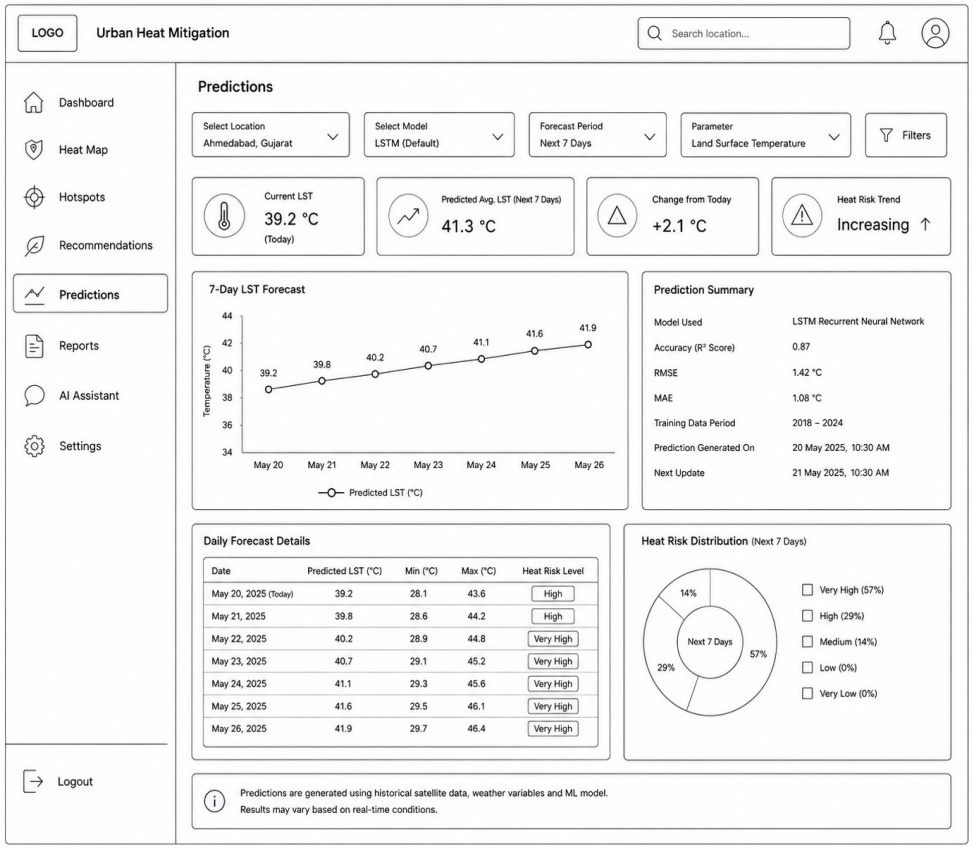


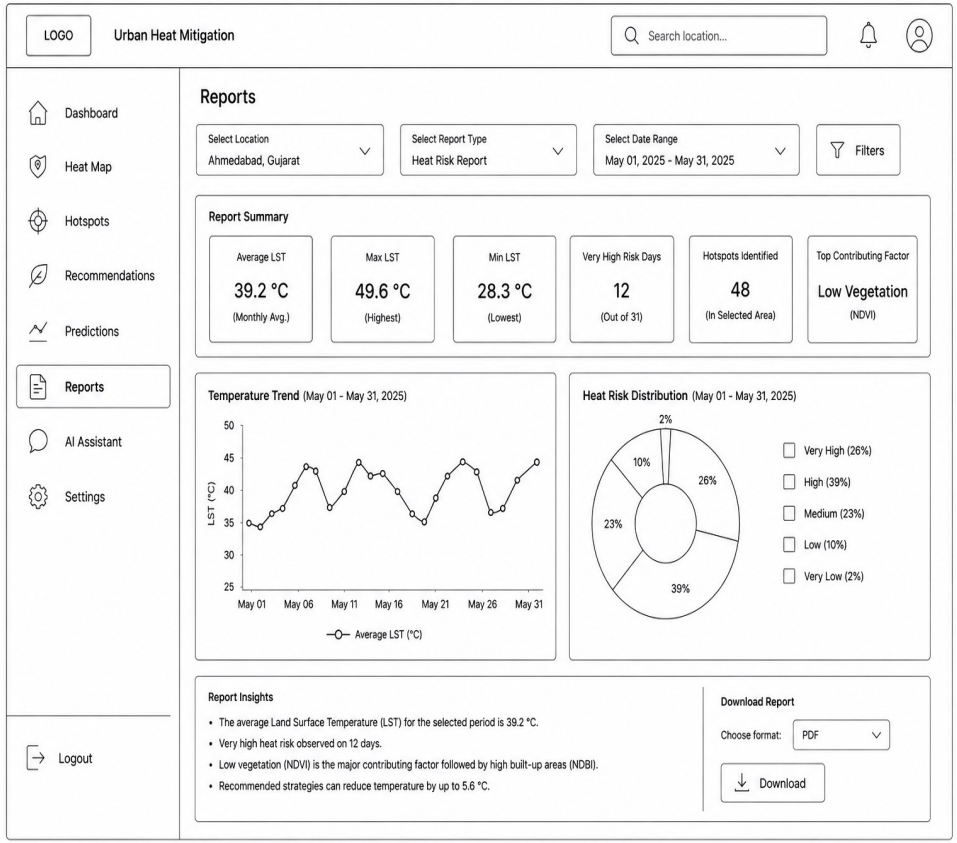
Wireframes/Mock diagrams of the proposed solution (optional)

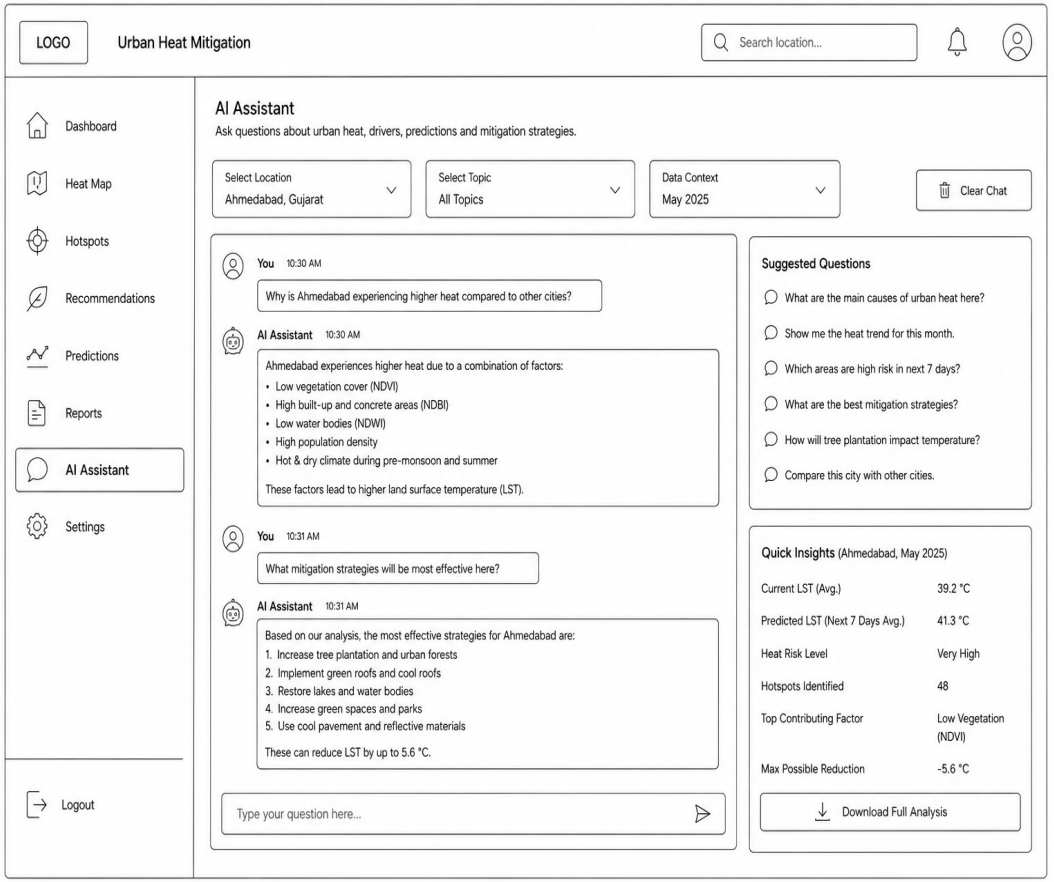


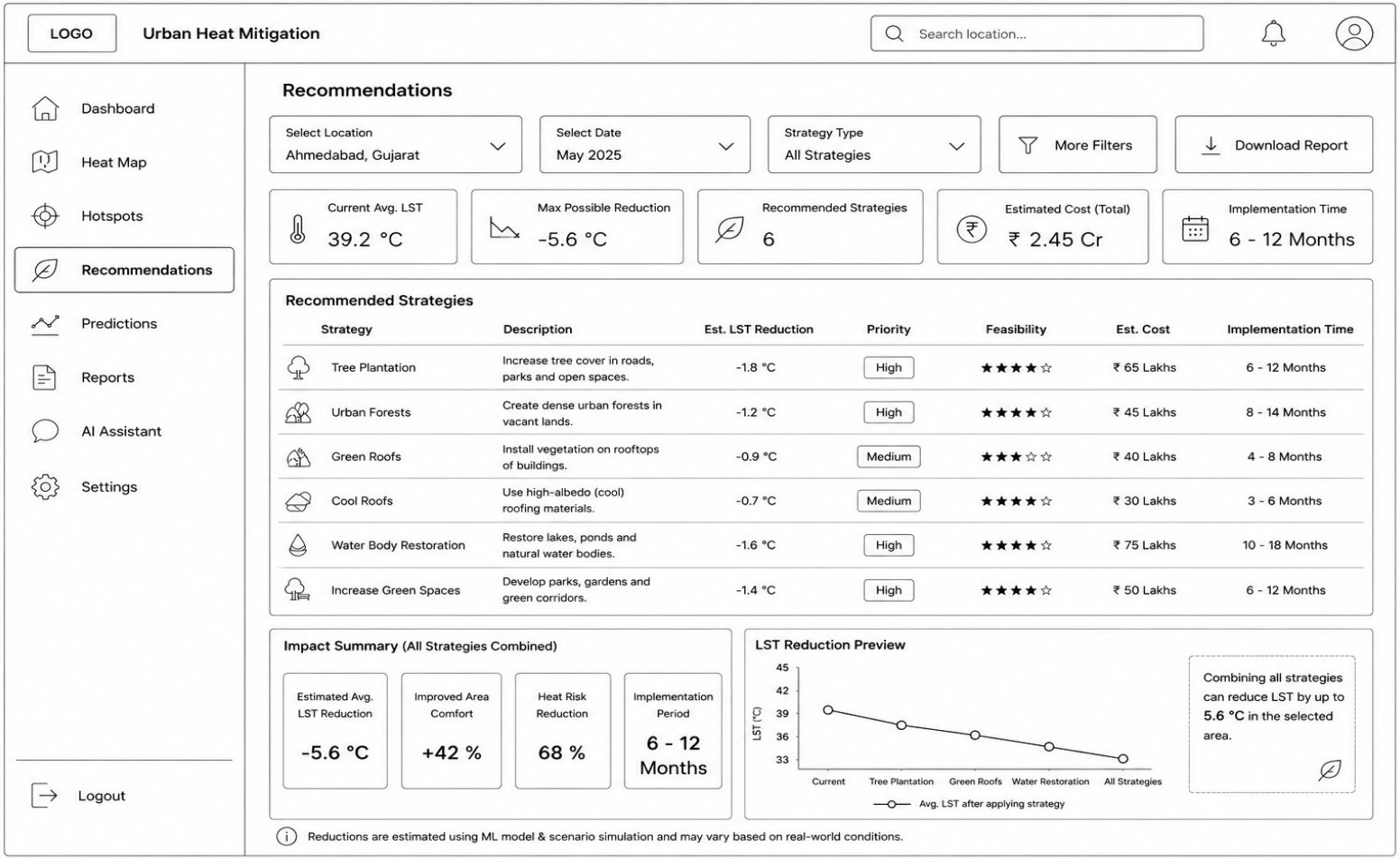



LOGO
Urban Heat Mitigation
Search location...
Dashboard
Heat Map
Hotspots
Recommendations
Predictions
Reports
AI Assistant
Settings
Logout
Hotspots
Select City
All Cities
Select Date
May 2025
Heat Risk Level
All
Filters
Total Hotspots
128
Very High Risk
48
High Risk
56
Medium Risk
18
Low Risk
6
Hotspots on Map
Heat Risk Level
Very High
High
Medium
Low
Top 10 Hotspots
Rank Location City Heat Risk LST (°C)
1 Chandkheda Ahmedabad Very High 49.6
2 Naroda Ahmedabad Very High 48.9
3 Mundra Kutch Very High 48.2
4 Jamnagar Jamnagar Very High 47.8
5 Delhi (Narela) Delhi Very High 47.6
6 Bhuj Kutch High 46.7
7 Jaipur (Sanganer) Jaipur High 46.3
8 Nagpur (Hingna) Nagpur High 45.9
9 Surat (Sachin) Surat High 45.7
10 Bhopal (MP Nagar) Bhopal High 45.5
Hotspot Details (Selected)
No Image Available
Location : Chandkheda
City : Ahmedabad, Gujarat
Heat Risk : Very High
LST (Now) : 49.6 °C
LST (Predicted – 7 Days) : 50.8 °C
Contributing Factors
Low Vegetation (NDVI) : 0.21 (Low)
High Built-up Area (NDBI) : 0.76 (High)
Low Water Bodies (NDWI) : 0.18 (Low)
Elevation : 53 m
Season : Pre-Monsoon
Recommended Actions
Increase tree plantation
Green roof implementation
Expand urban green spaces
Water body restoration
Cool roof materials
View Recommendations


LOGO
Urban Heat Mitigation
Search location...
Dashboard
Heat Map
Hotspots
Recommendations
Predictions
Reports
AI Assistant
Settings
Logout
Predictions
Select Location
Ahmedabad, Gujarat
Select Model
LSTM (Default)
Forecast Period
Next 7 Days
Parameter
Land Surface Temperature
Filters
Current LST
39.2 °C
(Today)
Predicted Avg. LST (Next 7 Days)
41.3 °C
Change from Today
+2.1 °C
Heat Risk Trend
Increasing
7-Day LST Forecast
Temperature (°C)
44
42
40
38
36
34
39.2
39.8
40.2
40.7
41.1
41.6
41.9
May 20
May 21
May 22
May 23
May 24
May 25
May 26
Predicted LST (°C)
Prediction Summary
Model Used LSTM Recurrent Neural Network
Accuracy (R² Score) 0.87
RMSE 1.42 °C
MAE 1.08 °C
Training Data Period 2018 – 2024
Prediction Generated On 20 May 2025, 10:30 AM
Next Update 21 May 2025, 10:30 AM
Daily Forecast Details
Date Predicted LST (°C) Min (°C) Max (°C) Heat Risk Level
May 20, 2025 (Today) 39.2 28.1 43.6 High
May 21, 2025 39.8 28.6 44.2 High
May 22, 2025 40.2 28.9 44.8 Very High
May 23, 2025 40.7 29.1 45.2 Very High
May 24, 2025 41.1 29.3 45.6 Very High
May 25, 2025 41.6 29.5 46.1 Very High
May 26, 2025 41.9 29.7 46.4 Very High
Heat Risk Distribution (Next 7 Days)
14%
Next 7 Days
57%
29%
Very High (57%)
High (29%)
Medium (14%)
Low (0%)
Very Low (0%)
Predictions are generated using historical satellite data, weather variables and ML model.
Results may vary based on real-time conditions.

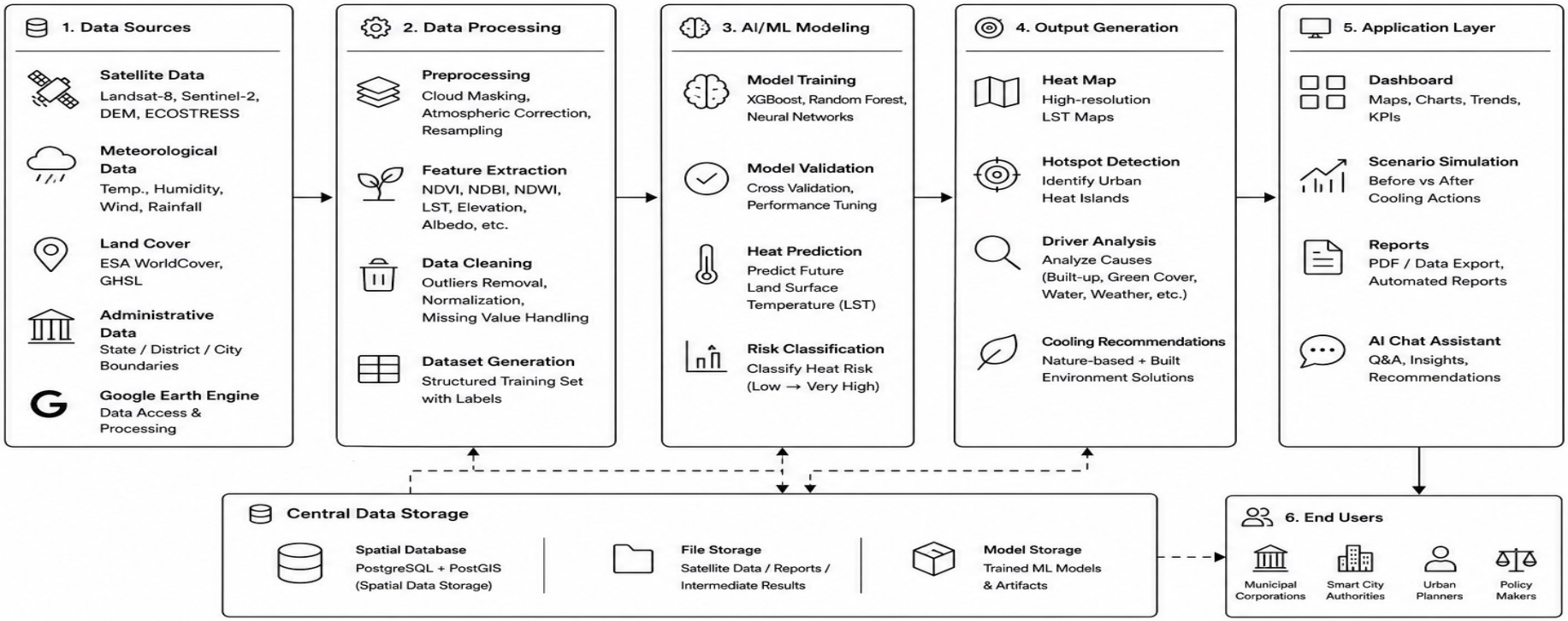

LOGO
Urban Heat Mitigation
Search location...
Dashboard
Heat Map
Hotspots
Recommendations
Predictions
Reports
AI Assistant
Settings
Logout
Reports
Select Location
Ahmedabad, Gujarat
Select Report Type
Heat Risk Report
Select Date Range
May 01, 2025 - May 31, 2025
Filters
Report Summary
Average LST
39.2 °C
(Monthly Avg.)
Max LST
49.6 °C
(Highest)
Min LST
28.3 °C
(Lowest)
Very High Risk Days
12
(Out of 31)
Hotspots Identified
48
(In Selected Area)
Top Contributing Factor
Low Vegetation
(NDVI)
Temperature Trend (May 01 - May 31, 2025)
LST (°C)
50
45
40
35
30
25
May 01
May 06
May 11
May 16
May 21
May 26
May 31
Average LST (°C)
Heat Risk Distribution (May 01 - May 31, 2025)
2%
10%
26%
23%
39%
Very High (26%)
High (39%)
Medium (23%)
Low (10%)
Very Low (2%)
Report Insights
• The average Land Surface Temperature (LST) for the selected period is 39.2 °C.
• Very high heat risk observed on 12 days.
• Low vegetation (NDVI) is the major contributing factor followed by high built-up areas (NDBI).
• Recommended strategies can reduce temperature by up to 5.6 °C.
Download Report
Choose format:
PDF
Download


LOGO
Urban Heat Mitigation
Search location...
Dashboard
Heat Map
Hotspots
Recommendations
Predictions
Reports
AI Assistant
Settings
Logout
AI Assistant
Ask questions about urban heat, drivers, predictions and mitigation strategies.
Select Location
Ahmedabad, Gujarat
Select Topic
All Topics
Data Context
May 2025
Clear Chat
You 10:30 AM
Why is Ahmedabad experiencing higher heat compared to other cities?
AI Assistant 10:30 AM
Ahmedabad experiences higher heat due to a combination of factors:
• Low vegetation cover (NDVI)
• High built-up and concrete areas (NDBI)
• Low water bodies (NDWI)
• High population density
• Hot & dry climate during pre-monsoon and summer
These factors lead to higher land surface temperature (LST).
You 10:31 AM
What mitigation strategies will be most effective here?
AI Assistant 10:31 AM
Based on our analysis, the most effective strategies for Ahmedabad are:
1. Increase tree plantation and urban forests
2. Implement green roofs and cool roofs
3. Restore lakes and water bodies
4. Increase green spaces and parks
5. Use cool pavement and reflective materials
These can reduce LST by up to 5.6 °C.
Type your question here...
Suggested Questions
What are the main causes of urban heat here?
Show me the heat trend for this month.
Which areas are high risk in next 7 days?
What are the best mitigation strategies?
How will tree plantation impact temperature?
Compare this city with other cities.
Quick Insights (Ahmedabad, May 2025)
Current LST (Avg.) 39.2 °C
Predicted LST (Next 7 Days Avg.) 41.3 °C
Heat Risk Level Very High
Hotspots Identified 48
Top Contributing Factor Low Vegetation (NDVI)
Max Possible Reduction -5.6 °C
Download Full Analysis


LOGO
Urban Heat Mitigation
Search location...
Dashboard
Heat Map
Hotspots
Recommendations
Predictions
Reports
AI Assistant
Settings
Logout
Recommendations
Select Location
Ahmedabad, Gujarat
Select Date
May 2025
Strategy Type
All Strategies
More Filters
Download Report
Current Avg. LST
39.2 °C
Max Possible Reduction
-5.6 °C
Recommended Strategies
6
Estimated Cost (Total)
₹ 2.45 Cr
Implementation Time
6 - 12 Months
Recommended Strategies
Strategy
Description
Est. LST Reduction
Priority
Feasibility
Est. Cost
Implementation Time
Tree Plantation
Increase tree cover in roads, parks and open spaces.
-1.8 °C
High
★★★★☆
₹ 65 Lakhs
6 - 12 Months
Urban Forests
Create dense urban forests in vacant lands.
-1.2 °C
High
★★★★☆
₹ 45 Lakhs
8 - 14 Months
Green Roofs
Install vegetation on rooftops of buildings.
-0.9 °C
Medium
★★★☆☆
₹ 40 Lakhs
4 - 8 Months
Cool Roofs
Use high-albedo (cool) roofing materials.
-0.7 °C
Medium
★★★★☆
₹ 30 Lakhs
3 - 6 Months
Water Body Restoration
Restore lakes, ponds and natural water bodies.
-1.6 °C
High
★★★★☆
₹ 75 Lakhs
10 - 18 Months
Increase Green Spaces
Develop parks, gardens and green corridors.
-1.4 °C
High
★★★★☆
₹ 50 Lakhs
6 - 12 Months
Impact Summary (All Strategies Combined)
Estimated Avg. LST Reduction
-5.6 °C
Improved Area Comfort
+42 %
Heat Risk Reduction
68 %
Implementation Period
6 - 12 Months
LST Reduction Preview
45
42
39
36
33
LST (°C)
Current
Tree Plantation
Green Roofs
Water Restoration
All Strategies
Avg. LST after applying strategy
Combining all strategies can reduce LST by up to 5.6 °C in the selected area.
Reductions are estimated using ML model & scenario simulation and may vary based on real-world conditions.

UrbanHeat AI — System Architecture

AI-Powered Urban Heat Mitigation & Cooling Recommendation Platform



Technologies to be used in the solution:

Frontend (User Interface)

- Used to build a fast, responsive, and interactive web application.
- React (Vite) – High-performance frontend framework
- Tailwind CSS – Responsive UI design
- React Leaflet & Leaflet – Interactive satellite and heat maps
- Axios – API communication
- Recharts – Charts and data visualization

Backend (Application Server)

- Handles business logic, APIs, authentication, and communication between the frontend and AI models.
- Node.js
- Express.js
- REST API
- JWT Authentication

Database

Stores environmental data, hotspot information, user data, and geospatial datasets.

- **PostgreSQL**
- **PostGIS** (Spatial Database Extension)

Artificial Intelligence & Machine Learning

- Used for heat prediction, hotspot detection, explainable AI, and decision support.
- Python
- PyTorch
- Scikit-learn
- XGBoost
- SHAP (Explainable AI)

Geospatial & Remote Sensing

- Processes satellite imagery and environmental datasets for heat analysis.
- Google Earth Engine
- Rasterio
- GDAL
- GeoPandas
- Shapely
- Xarray
- NumPy
- Pandas

Data Sources

➢ Satellite and environmental datasets used for model training and heat analysis.

- Landsat 8 (Land Surface Temperature)
- ECOSTRESS
- Sentinel-2
- ERA5 Weather Data
- OpenStreetMap
- Global Human Settlement Layer (GHSL)

Visualization

➢ Displays heat maps, trends, predictions, and analytics through interactive visualizations.

- Leaflet Maps
- Recharts
- Plotly

APIs & Data Processing

- Collects, processes, and manages environmental and geospatial data.
- Google Earth Engine API
- OpenStreetMap API
- Python ETL Scripts

Development Tools

- Used for coding, testing, version control, and project management.
- Git
- GitHub
- Docker
- Postman
- Jupyter Notebook

Deployment

- Hosts the complete application on cloud platforms.
- Vercel (Frontend)
- Render (Backend)
- Supabase PostgreSQL (PostGIS)

Future Enhancements

- Advanced tools for optimization, simulation, and urban climate modeling.
- Optuna
- DEAP (Genetic Algorithms)
- SOLWEIG
- InVEST

Estimated implementation cost Around (optional):

• **Cloud Infrastructure**	• Virtual server, database, and cloud storage (3 months)	• ₹3,186
• **AI/ML Model Training**	• Cloud GPU/CPU resources for model training	• ₹2,478
• **Web & API Deployment**	• Frontend, backend, and API hosting	• ₹1,364
• **Domain & SSL Security**	• Domain registration and SSL certificate	• ₹978
• **Data Storage & Backup**	• Secure storage and periodic backups	• ₹1,129
• **Testing & Quality Assurance**	• Performance testing, debugging, and monitoring	• ₹1,189
• **Total Estimated Cost**	• **Pilot Prototype Deployment**	• **₹10,324**

Kaggle Dataset Link

India Environmental Dataset (2021–2025)

https://www.kaggle.com/datasets/vishalbaraiya1014/india-environmental-dataset-2021-2025

GitHub Repository Link

https://github.com/aniruddh209/Urban-Heat-Mitigation-IsroHecathon


इसरो isro
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HACK2SKILL
BHARATIYA
ANTARIKSH
HACKATHON
2026
THANK YOU